



Progressive reinforcement learning for video summarization

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ARTICLE INFO

Keywords:

Video summarization

Progressive reinforcement learning

Hierarchical strategy

ABSTRACT

Video summarization addresses generating video summaries to help watchers grasp the content of a video without watching it entirely. Many methods have engaged in automatic video summarization. Although these methods have performed well, they still suffer from limited training data and sparse reward problems. We propose a Progressive Reinforcement Learning Video Summarization structure (PRLVS) with an unsupervised reward. The reward measures the information and quality the selected frames convey without annotations. Striving to earn higher rewards, our PRLVS adopts a “T”-type human thinking paradigm: choosing some key frames and checking if their adjacent frames are better than them. To simulate this paradigm, we decompose the flat strategy into a hierarchical strategy consisting of a horizontal policy and a vertical policy. These two policies are optimized alternatively, which densifies the reward while reducing the exploration space. Their cooperation also makes the agent capture the context information of the whole video at every step. Extensive experimental results on two benchmark databases (i.e., SumMe, TVSum) show that our PRLVS outperforms the comparisons and approaches the supervised methods, which indicates that it is significant to integrate our unsupervised reward into the progressive reinforcement learning structure to address limited annotation and sparse reward problems.

1. Introduction

In recent years, video-on-demand (VOD) applications have shown colossal market value. Many VOD platforms (e.g., YouTube, TikTok) enjoy great popularity and bring substantial economic benefits. A hefty profit urges video uploaders to endeavor to drive traffic to their videos. The first thing that occurs to those uploaders is selecting some representative frames as a summary [1–3]. A good summary is useful. Not only does it help uploaders to gain clicks and followers, but also help users to improve browsing efficiency. Therefore, automatic video summarization is regarded as a remedy for serving both uploaders and users [4].

Numerous methods have been proposed in the literature. In general, they can be divided into two directions: 1) A paradigm selects key frames based on video segmentation [5–7]. The segment positions are the change points of the importance score curve. 2) A paradigm that directly identifies a set of key frames [8,9]. To provide global supervision available at the end of the key frame sequence, Reinforcement Learning is inherently suitable [10,11]. Though these methods have achieved high performance, they still need some help with some problems.

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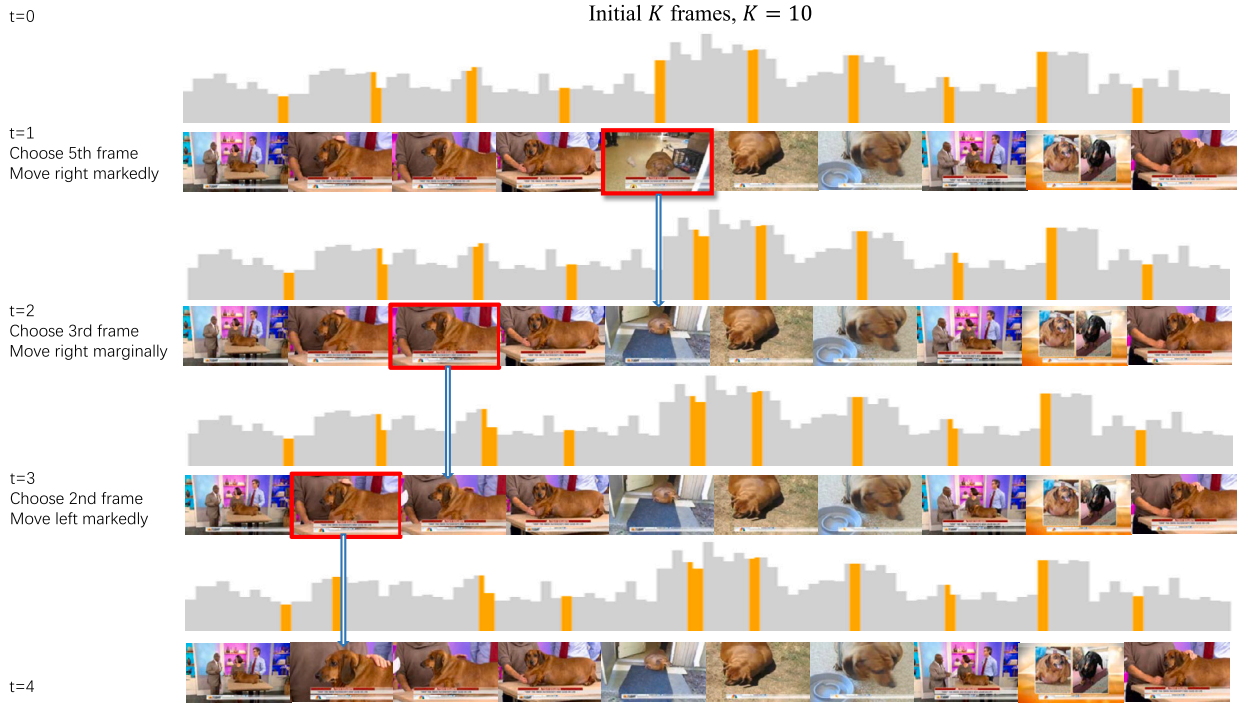


Fig. 1. An example of how our PRLVS summarizes a video by simulating the horizontal-vertical thinking paradigm. The red bounding box denotes the horizontal thinking schema: choose which frame to update. “move right markedly”, “move right marginally”, and “move left markedly” denote the vertical thinking schema: how to change this frame. The images are frames selected at the current step. The gray bin denotes the ground-truth important score of all frames in the entire video, while the orange bin denotes that of selected frames.

In the first paradigm, the change points in the important score curve are sensitive to the noise [12]: a method may neglect some change points with too “tight” a constraint of the number of changes and otherwise would include noise into change points. The research focus in the literature has gradually shifted to the second paradigm to avoid the rigid number setting.

In the second paradigm, The sparse reward problem leads to hard convergence [10] as an agent receives the reward only after the whole summary is generated in conventional reinforcement learning methods. Moreover, models suffer from limited training data as it is tedious to label every frame by importance score, with the result that few databases are available for supervised learning.

To address the problem of sparse reward and data limitation, we propose a Progressive Reinforcement Learning network for Video Summarization (PRLVS) in an unsupervised learning fashion. In our PRLVS, we decompose the summarization task hierarchically. The decomposition is inspired by the “T”-type thinking paradigm in human cognition mechanism [13,14]: people always select useful information from a video in a holistic perspective and then search in-depth. Namely, people will select key frames as an intermediate summary representing the video. To obtain a better summary beyond the current intermediate summary, they iteratively pick a key frame, and determine if any of its adjacent frames are superior to the selected frame, repeating this process until forming the final summary. The key frame chosen in an iteration is termed anchor.

Our model aims to train an agent to modify the summary progressively. The agent chooses to replace one frame with some frame in the neighborhood and receives a reward for the whole summary at each step, not just at the last step. While alleviating the sparse reward problem, this strategy also enlarges the search space. The challenge is reducing the search trials in the expanded space.

To improve the search efficiency, we model the “T”-type thinking paradigm into an agent by implementing two policies: 1) The horizontal policy contributes to sampling coarse summaries from raw videos. 2) The vertical policy contributes to refining the sampled frames to enrich semantic information with minimal frames. We depict an example in Fig. 1 to show the concrete process of how PRLVS completes this task. As shown in the figure, the agent chooses the fifth selected frame as the anchor and moves it right markedly when $t = 1$. In the next step, we can find that the fifth frame is changed to an image with a large semantic difference with the image at step $t = 1$. The whole process is completed by optimization of two policies.

We alternatively iterate a horizontal policy or a vertical policy in model optimization. In an iteration, the vertical policy directly interacts with the external environment with a reward measuring representativeness and diversity. Then the reflections are transferred to the horizontal policy. As for the horizontal policy, we design an unsupervised reward measuring the amount of information conveyed by summaries. Through this reward, the horizontal policy summarizes the reflections and leverages overall semantic features to update frame selection.

Extensive experiments show that our method outperforms the comparisons on two commonly used databases in video summarization: SumMe [1] and TVSum [15]. The results also exhibit the proposed PRLVS can (i) efficiently discover hierarchical information

rather than smooth expression; (ii) provide a stable policy that is insensitive to the scale of the overall selection. The contribution can be listed as follows.

- We simulate the human thinking process when picking frames from videos by designing a reinforcement learning structure.
- We design an unsupervised reward function based on information theory and integrate it into the progressive reinforcement learning framework.
- We optimize an agent by alternatively updating a horizontal policy or a vertical policy to summarize hierarchical information progressively.
- Experimental results show the proposed method PRLVS outperforms the comparisons.

This paper is organized as follows. Section 2 describes the related work. The proposed method and experimental results are presented in Section 3 and Section 4, respectively. The conclusion and future work are presented in Section 5.

2. Related work

2.1. Video summarization

Video summarization is formulated as sampling a set of key frames from a video [4,5,16]. Intuitively, methods [17,6,18,19] can learn video summarization supervised by the annotated frame-level score or shot-level score of raw videos. These supervised methods learn a visual representation further to derive compact and representative video summaries. The visual representation should encode temporal dependency among video frames. Recently, the encoder has been implemented by Neural Networks (e.g., LSTM [8]). In addition to annotated scores, semantic side information can be included for encoding. Zhang et al. [7] extracted highlight subsets by transferring video category information to input videos. More recently, DSNet [20] is proposed to detect interest as a plug-in. Though supervised methods achieve high performance, few well-prepared databases have been proposed in the form of videos and their corresponding summaries. Therefore, weakly supervised and unsupervised methods are also hot in the research community.

Weakly supervised methods introduce video-level semantic information to compensate for model training [21,11]. Unsupervised methods [22,5,12,9] always depend on a well-designed criterion, not the annotations. Song et al. [15] utilized title-based image search results to generate a summary in an unsupervised style. Zhou et al. [10] designed a reward including diversity and representativeness. Without adopting criteria to bridge the summary and the video, Mahasseni et al. [23] directly compared the generated summary and the original video by designing a discriminator in generative adversarial networks (GANs). Recently, some work combined LSTM and GCN to learn the frame-level and shot-level dependencies segment-by-segment [24]. Without the limitation of annotation, unsupervised methods can introduce more raw data or another task. Rochan et al. [25] proposed an unpaired strategy by separately organizing the video database and summary database. Jiang et al. [26] combined video summarization and video moment retrieval. Well-designed criteria and large-scale raw data make the performance of the unsupervised methods approach that of the supervised ones.

In this work, we adopt an unsupervised fashion and compare our PRLVS with these three types of methods.

2.2. Reinforcement learning

Reinforcement Learning has been widely applied in the video understanding literature [27]. This can help researchers train their model fully automatically and keep training and test processes in the same setting. For example, the video caption model has always been trained by cross-entropy loss, while evaluated on phrase-matching-based metrics. To bridge the gap, a reinforcement learning structure with a reward measuring sentence-level phrase-matching metrics is proposed [28]. In many other video understanding tasks, reinforcement learning can be used to implement different modules such as feature aggregation [29].

As for video summarization, reinforcement learning has also played a significant role in frame selection. Zhou et al. [10] introduced reinforcement learning to pick out representative frames directly. In contrast, Lan et al. [30] applied a policy skipping irrelevant video frames. Despite the frame-level policies, some researchers design shot-level policies. Lei et al. [31] parsed a video into action instances and designed a reinforcement learning model to select the most distinguishable instances, which can weak annotation's ambiguity. In sum, the reward is defined globally on a whole summary, while the actions are designed to pick one single frame. This difference results in the sparse reward problem. To compensate for the sparse reward, Chen et al. [32] proposed a hierarchical reinforcement learning method with designed sub-rewards. Though this strategy resolved the sparse problem, it shrank the search space and neglected the context.

In this work, we design a progress reinforcement learning model to encode context while maintaining search efficiency.

3. Proposed method

In this work, we formulate key frames selection as a Markov Decision Process (MDP) and propose a progressive reinforcement learning model to construct the interactions between agents and videos. As discussed above, the two major limitations of the current video summarization methods are 1) limited data problem: the ground-truth important scores are limited due to the lack of training data; 2) sparse reward problem: a flat policy can only calculate global reward one time on the whole summary, which results in hard convergence. To address these two problems, we train an agent to simulate the "T"-type thinking process with a horizontal policy

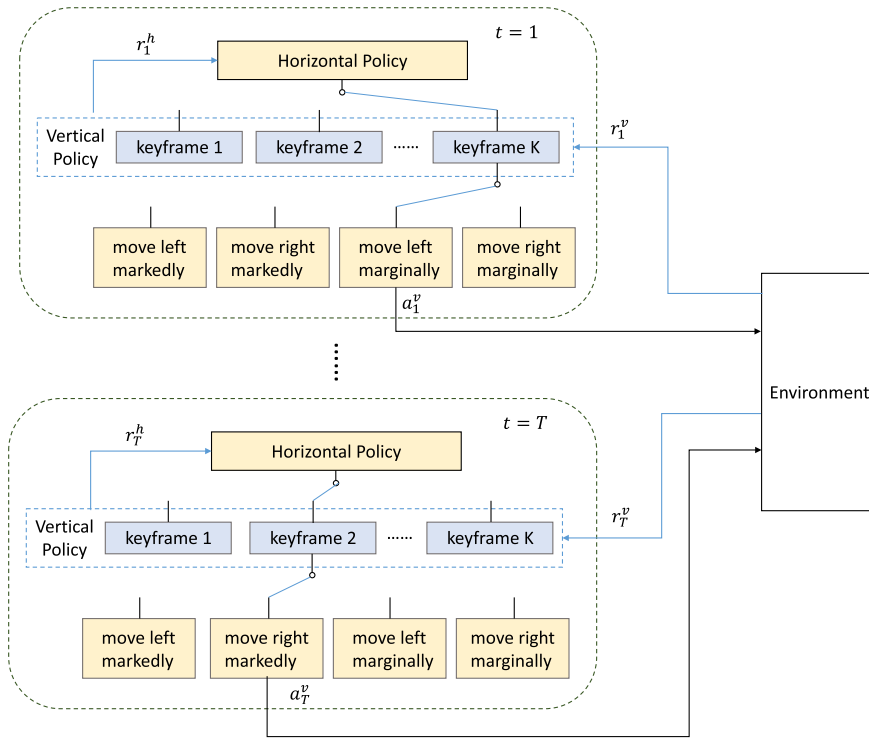


Fig. 2. A illustration of the working pipeline of the horizontal policy and vertical policy.

and a vertical policy: the horizontal policy is trained to choose some frame to update, and the vertical policy is trained to change the selected frames.

3.1. Problem formulation

The pipeline of our proposed PRLVS is shown in Fig. 2. Let N be the frame length of the entire video. The agent samples K frames as intermediate summary at each step. Let $v = \{v_1, \dots, v_N\}$ be one video in the database and v_i is the i_{th} frame of v . Consequently, we use $v^k = \{v_1^k, \dots, v_K^k\}$ to denote the intermediate summary, where v_j^k is the j_{th} key frame. We also design a mapping function $g(j)$ to map the j -th key frame to the original index in video v . The video summarization process is denoted by a tuple $\langle s, \langle a^h, a^v \rangle, \mathcal{T}, r \rangle$. $s \in S$ denotes a state of the agent, $\langle a^h, a^v \rangle \in \mathcal{A}$ denotes an action tuple consisting of horizontal action and vertical action, $\mathcal{T} : (s, \langle a^h, a^v \rangle) \rightarrow s'$ denotes the transition function, and r denotes the reward at some step.

State. The extracted features f of K frames at t is the state s_t . Inspired by the segmentation paradigm [12], we regard a frame as the most fine-grained video segmentation and use the common feature extractor ϕ_v to extract features from frames. During the training, we keep the selected key frames following the temporal order. When the order is broken, a termination signal is sent. After termination, we do not need to reorganize the selected frames. Therefore, we focus on the total information rather than the temporal semantic relationship among frames. As for the size of intermediate summary K , it is similar to the number of agents in the multi-agent learning scenario, where each frame is regarded as an agent. In this scenario, the number of agents has little influence on efficacy [33]. In our PRLVS, we will analyze the influence of K in the following section.

Action. We use the action tuple $\langle a^h, a^v \rangle$ to formulate the hierarchical selection process. The action a^h is used to select a specific frame as an anchor, which is the entrance of the hierarchical action space. Each selected frame is a semantic branch for further movement executed by a^v .

a^v is classified into four classes: move left markedly, move right markedly, move left marginally, and move right marginally. 1) ‘move left markedly’ and ‘move right markedly’ are symmetry actions for choosing a frame τ^l steps from the selected frame. τ^l is fixed to $N/50$, where N denotes the number of total frames in the entire video. These two actions simulate a case in which a person may think the current frame is far from his expectation and will try to find another one by moving a large step. 2) ‘move left marginally’ and ‘move right marginally’ are also symmetry actions for choosing a frame τ^s from the selected frame, and τ^s is set to 1. These two actions simulate a case in which a person may think a frame i is close to his expectation but still try to find a better one in the neighborhood of i .

Policy. We design a “T”-type policy to train the agent to simulate the human horizontal-vertical thinking paradigm when summarizing videos. We decompose complex video summarization via a two-stage progressive manipulation. This can better guarantee

convergence than using a flat policy to sample key frames according to an overall reward iteratively. Just as the name implies, “T”-type policy consists of a horizontal policy and a vertical policy. At each step, the horizontal policy $\pi^h(a_t^h|s_t)$ determines which frame will be selected as an anchor. The vertical policy $\pi^v(a_t^v|s_t, a_t^h)$ determines which operation will be adopted to update the current frame.

The horizontal policy targets to choose an entrance to modify the intermediate summary and guide the following vertical policy according to the following scenarios: 1) The selected frame i cannot rank as key frames, a temporally adjacent frame whose index is in $\{i - \tau^l, i + \tau^l\}$ can be ranked, where the agent should use “move left/right markedly policy”. 2) The selected frame i can reflect a short activity, but an adjacent frame can convey more information than this one, where the agent should use “move left/right marginally”. Concretely, at step t , the “T”-type policy firstly samples a_t^h from horizontal policy π^h to choose the frame to edit:

$$a_t^h \sim \pi^h(a_t^h|s_t) \quad (1)$$

Then an atomic action is sampled from vertical policy π^v to update the frame:

$$a_t^v \sim \pi^v(a_t^v|s_t, a_t^h) \quad (2)$$

3.2. Progressive reinforcement learning

Reward. Our PRLVS aims at selecting informative frames with fewer steps without supervised labels. Therefore, we choose a function encouraging richer information as the reward function rather than cross-entropy or F1-score. Concretely, we design two rewards for vertical and horizontal policies, respectively. The reward for the vertical policy is r^v , while the reward for the horizontal policy is r^h .

The vertical reward r^v aims at guiding the agent to generate diverse and representative summaries. It is similar to the diversity-representativeness reward [10]. r^v is computed by directly interacting with the environment. It consists of two parts: 1) r_d^v measures the diversity among selected frames; 2) r_r^v measures the representativeness of selected frames.

For r_d^v , we calculate the pairwise dissimilarity between any two selected frames. r_d^v of the intermediate summary $\mathbf{v}^k$ can be denoted as:

$$r_d^v = \frac{1}{K(K-1)} \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K (1 - \cos(\mathbf{x}_i^k, \mathbf{x}_j^k)) \quad (3)$$

where $\mathbf{x}_i^k$ and $\mathbf{x}_j^k$ are feature vectors of frame $\mathbf{v}_i^k$ and $\mathbf{v}_j^k$, respectively. $\cos(\cdot, \cdot)$ denotes the cosine similarity.

Intuitively, the diversity reward works in the case that two frames are temporally close. Its effect will be weakened if the two feature vectors exhibit large dissimilarity while the corresponding selected frames are temporally far from each other. Therefore, r_d^v is set to 1 if $|g(j) - g(i)| > 2$.

For r_r^v , we consider the representativeness as the minimal distance between a selected frame and its nearest cluster center point.

$$r_r^v = \exp\left(-\frac{1}{K} \sum_{j < K} \min \| \mathbf{x}_i^k - \mathbf{x}_j^k \|_2\right) \quad (4)$$

To construct the final r^v , we combine r_d^v and r_r^v by adding them together.

$$r^v = 0.5 * r_d^v + 0.5 * r_r^v \quad (5)$$

The horizontal reward r^h aims at providing high-level comprehension by choosing some frame for the following actions. As the horizontal policy does not interact with the environment directly, r^h should be computed by accumulating the information from the vertical policy. To construct r^h , we adopt a common rule in Information Theory that the less likely an event is, the richer the message it conveys. Inspired by this rule, we design r^h by estimating the product of probabilities selecting a specific frame.

I_t is the amount of information at the t step, which is denoted by the product of marginal probabilities.

$$I_t = \prod_{i=1}^K p(a_i^h = i | s_t) \quad (6)$$

As the horizontal policy interacts with the environment through the vertical policy, r^h reflects the influence of the following steps executed by selecting a frame.

$$r^h = I_t - I_{t-1} \quad (7)$$

Optimization We use A2C [34] structure to learn the horizontal policy $\pi^h(a_t^h|s_t)$ and vertical policy $\pi^v(a_t^v|s_t, a_t^h)$. These two policies are separate and have different value functions, i.e., $V^h(s_t)$ and $V^v(s_t, a_t^h)$. Initially, the agent randomly selects K frames as the initial intermediate summary. The iterative process is visualized in Fig. 2.

Given a trajectory $\tau = \langle s_t, \pi^h(a_t^h|s_t), a_t^h, r_t^h, \pi^v(a_t^v|s_t, a_t^h), a_t^v, r_t^v \rangle, t = \{1, \dots, T\}$, we use R_t^h and R_t^v to denote the accumulated reward of horizontal policy and vertical policy R_t^v , respectively:

Algorithm 1: Optimization of PRLVS

input : discount factor γ , learning rate η , reward functions r^h and r^v , max time step T , total number of iterations E_i , number of iterations each policy is trained in a cycle E_c , the size of intermediate summary K , the optimizer opt

output: model parameters θ

```

1 initialize  $\theta$ ;
2 for  $i \leftarrow 0$  to  $E_i$  do
3   for  $j \leftarrow 0$  to  $B$  do
4     extracting a batch of data;
5     if  $j \% E_c == 0$  then
6       policy = Not (policy);
7       if policy == horizontal then Freeze (vertical);
8       else Freeze (horizontal);
9     end
10    for  $t \leftarrow 0$  to  $T$  do
11      if  $t == 0$  then initialize  $s$  from environment;
12      perform  $a_t^h$  by  $\pi^h(a^h|s_t)$ ;
13      get  $s^{t+1}$  and  $r_t^h$  from environment;
14      if policy == vertical then
15        perform  $a_t^v$  by  $\pi^v(a^v|s_t, a_t^h)$ ;
16        get  $s^{t+1}$  and  $r_t^v$  from environment;
17      end
18    end
19    if policy == horizontal then
20      calculate  $R_t^h$  according to Eqn. (8);
21      calculate loss for actor  $\mathcal{L}^h(\theta_{\pi^h})$  according to Eqn. (9);
22      calculate loss for critic  $\mathcal{L}^h(\theta_{V^h})$  according to Eqn. (11)
23    else
24      calculate  $R_t^v$  according to Eqn. (8);
25      calculate loss for actor  $\mathcal{L}^v(\theta_{\pi^v})$  according to Eqn. (10);
26      calculate loss for critic  $\mathcal{L}^v(\theta_{V^v})$  according to Eqn. (12)
27    end
28     $opt.zero\_grad()$ ;
29     $opt.step()$ ;
30  end
31 end

```

$$\begin{aligned}
R_t^h &= \begin{cases} r_t^h + \gamma V^h(s_t) & t = T \\ r_t^h + \gamma R_{t+1}^h & t = 1, \dots, T-1 \end{cases} \\
R_t^v &= \begin{cases} r_t^v + \gamma V^v(s_t, a_t^h) & t = T \\ r_t^v + \gamma R_{t+1}^v & t = 1, \dots, T-1 \end{cases}
\end{aligned} \tag{8}$$

where T is the max time step and γ is the discount factor.

With the formulated reward function, our PRLVS optimizes the agent in a mini-batch fashion [35]. We design two separate loss functions for the actor and critic, respectively.

For the actor, our PRLVS trains the agent by maximizing the policy gradient $\mathcal{L}^h(\theta_{\pi^h})$ and $\mathcal{L}^v(\theta_{\pi^v})$:

$$\mathcal{L}^h(\theta_{\pi^h}) = -\frac{1}{B} \sum_{b=1}^B \sum_{t=1}^T [\log \pi^h(a_t^h|s_t)(R_t^h - V^h(s_t))] \tag{9}$$

$$\mathcal{L}^v(\theta_{\pi^v}) = -\frac{1}{B} \sum_{b=1}^B \sum_{t=1}^T [\log \pi^v(a_t^v|s_t, a_t^h)(R_t^v - V^v(s_t, a_t^h))] \tag{10}$$

where B is the batch size of mini-batch. θ_{π^h} and θ_{π^v} are the parameters of policy networks.

For the critic, our PRLVS optimizes the agent by minimizing the squared difference between the accumulated reward and the value.

$$\mathcal{L}^h(\theta_{V^h}) = \frac{1}{B} \sum_{b=1}^B \sum_{t=1}^T (R_t^h - V^h(s_t))^2 \tag{11}$$

$$\mathcal{L}^v(\theta_{V^v}) = \frac{1}{B} \sum_{b=1}^B \sum_{t=1}^T (R_t^v - V^v(s_t, a_t^h))^2 \tag{12}$$

where θ_{V^h} and θ_{V^v} are the parameters of value networks.

The optimization of the two policies is an alternative process: 1) when training the horizontal policy, the vertical policy is fixed; 2) when training the vertical policy, the horizontal policy is fixed. An algorithmic description of the optimization is given in Algorithm 1.

Table 1
Main statistics of databases.

database	Video Source	No. of Videos	Annotation Type
SumMe [1]	User videos	25	Frame-level score
TVSum [15]	Web videos	50	Frame-level score
YouTube [22]	Web videos	39	key frame set
OVP ¹	Various genre videos	50	key frame set

Let E_t and E_c denote the total number of iterations and the number of iterations each policy is trained in a cycle. Following [36], we define the balance factor λ as:

$$\lambda = \lfloor \frac{E_t}{E_c} \rfloor \mod 2 \quad (13)$$

The final loss function is defined by combining the loss in Equation (9), (11), (10), (12), (13):

$$\mathcal{L} = \lambda(\mathcal{L}^h(\theta_{\pi^h}) + \mathcal{L}^v(\theta_{\pi^v})) + (1 - \lambda)(\mathcal{L}^h(\theta_{\nu^h}) + \mathcal{L}^v(\theta_{\nu^v})) \quad (14)$$

Termination The conventional reinforcement learning methods always include a termination signal as an additional action, which relies on accurate label annotations. Without annotations in our unsupervised manner, we design a “keep-order” setting to terminate the model automatically.

The “keep-order” setting is formulated as a rule that the first selected frame should always be temporally earlier than the second selected frame. To ensure the order of the selected K frames, we check whether the order is changed after a_v execution. We set a state *terminate* if the order is changed:

$$\begin{cases} s_t = \text{terminate} & \text{if } \#(g(j) > g(j+1)) > 0 \\ \text{not terminate} & \text{otherwise} \end{cases} \quad (15)$$

where $\#(\cdot)$ return the number and j denote j -th selected key frame.

4. Experiment

4.1. Database

We evaluate our method on two standard video summarization databases: SumMe [1] and TVSum [15]. We also exploit YouTube [22] and the OVP¹ to augment training data following the settings in [8]. The main characteristics of the databases are shown in Table 1.

4.2. Evaluation metrics

F-score: Given a test video v , the most common evaluation metric to compare the generated summary (A) and the ground truth (B) is F-score (F). The precision (P) and recall (R) are defined as follows:

$$P = \frac{A \cap B}{A}, R = \frac{A \cap B}{B} \quad (16)$$

$$F = \frac{2 \times P \times R}{P + R} \quad (17)$$

Rank correlation coefficients: This metric is proposed in [37] to evaluate the performance considering rank order statistics. In this work, we choose the same metrics Kendall's τ [38] and Spearman's ρ [39] correlation coefficients.

4.3. Implementation details

We choose GoogleNet [40] as the feature extractor ϕ_v , which is consistent with the comparisons. The size of the feature vector is 1024. The size of the hidden feature in the LSTM is 256. We set “move left/right markedly” to move five steps and set “move left/right marginally” to move one step. The size of intermediate summary K is set to 10. During the training stage, the switch iteration E_c is set to 10. We use Adam as the optimizer with a learning rate set to 1e-05 and weight decay set to 1e-05. The batch size is set to 1, and the maximal number of learning epochs is set to 300.

4.4. Comparison

In this section, we compare our PRLVS with some state-of-the-art baselines:

¹ Open video project. <https://open-video.org/>.

Table 2

Quantitative comparison with the competitors in terms of F-score (%). FS, WS, and US are short for fully-supervised or unpaired, weakly-supervised, and unsupervised methods, respectively. The best performance in unsupervised methods is in **bold** and the second-best performance is underlined.

Supervision	Method	SumMe	TvSum
FS	Interesting	39.4	–
	Submodular	39.7	–
	Summary transfer	40.9	–
	vsLSTM	37.6	54.2
	dppLSTM	38.6	54.7
	SUM-GAN _s	41.7	56.3
	DR-DSN _s	42.1	58.1
	Retrospective	44.9	63.9
	FCSN	47.5	56.8
	DSNet-AB	50.2	62.1
	DSNet-AF	51.2	61.9
	RSGN	45.0	60.1
	iPTNet	54.5	63.4
WS	Unpaired-GAN	47.5	55.6
	HRL	43.6	58.4
	VCSN	41.5	55.7
US	Vsumm	33.7	–
	TVSum	–	50.0
	DR-DSN	41.4	57.6
	FCSN _u	41.5	52.7
	Less is more	–	56.3
	1 Minute	45.6	<u>59.1</u>
	ACGAN	<u>46.0</u>	58.5
	RSGN _{uns}	42.3	58.0
	PRLVS (ours)	46.3	63.0

Table 3

Quantitative comparison with methods in terms of rank correlation coefficients. The best performance in unsupervised methods is in **bold** and the second-best performance is underlined.

Supervision	databases	SumMe		TVSum	
		τ	ρ	τ	ρ
FS	dppLSTM	0.071	0.101	0.042	0.055
	DSNet-AB	0.051	0.059	0.108	0.129
	DSNet-AF	0.037	0.046	0.113	0.138
	RSGN	0.083	0.085	0.083	0.090
	iPTNet	0.101	0.119	0.134	0.163
US	DR-DSN	0.011	0.014	0.020	0.026
	RSGN _{uns}	<u>0.071</u>	<u>0.073</u>	0.048	<u>0.052</u>
	PRLVS (ours)	0.075	0.089	0.080	0.131
human		0.205	0.213	0.177	0.204

Full-Supervised Methods: Interesting [1], Submodular [6], Summary transfer [7], vsLSTM [8], dppLSTM [8], SUM-GAN_s [23], DR-DSN_s [10], Retrospective [41], FCSN [19], DSNet-AB [20], DSNet-AF [20], RSGN [24], iPTNet [26], Unpaired-GAN [25].

Weakly-Supervised Methods: HRL [32], VCSN [11].

Unsupervised Methods: Vsumm [22], TVSum [15], DR-DSN [10], FCSN_u [19], Less is more [42], 1 Minute [43], ACGAN [44], RSGN_{uns} [24].

We compare our PRLVS with the competitors in F-score (Table 2), rank correlation coefficients (Table 3), and its variants (Table 4). To further analyze our PRLVS, we also validate the methods in the Augmented (A) and Transfer (T) settings, following [10] (Table 5). Compared to the F-score competitors, our PRLVS outperforms unsupervised and weakly supervised methods and approaches the supervised method with the best performance. When compared in terms of rank correlation coefficients, our PRLVS is the closest to human assessment.

Table 4

Comparison between PRLVS and PRLVS's variants (PRLVS-a, PRLVS-s, and PRLVS-f).

K	SumMe			TVSum		
	F-score	τ	ρ	F-score	τ	ρ
PRLVS-a	46.9	0.08	0.10	63.1	0.10	0.14
PRLVS-s	47.6	0.01	0.07	62.0	0.07	0.10
PRLVS-f	42.7	0.02	0.02	58.6	0.03	0.05
PRLVS	46.3	0.07	0.09	63.0	0.08	0.13

4.4.1. Comparison in terms of F-score

Comparison with supervised approaches. On SumMe, Our PRLVS approaches the top method, iPTNet (85%). As On TVSum, Our PRLVS performs closely to the top method, Retrospective (99%). For Unpaired-GAN [25], it leverages the additional databases (i.e. OVP and Youtube) during the training phase, while our PRLVS only uses the given database. This result validates the efficacy of our progressive learning structure.

Comparison with weakly supervised approaches. Our PRLVS leads the competitors on both SumMe and TVSum by a large margin (6.2%-11.6% on SumMe and 7.9%-13.1% on TvSum). HRL [32] decomposes the agent by designing sub-workers to achieve sub-goals independently. The sub-workers can transfer information from the environment, while the agent cannot interact with the environment directly. However, the agent in our PRLVS has an overall understanding of the environment through the combination of horizontal and vertical policies.

Comparison with unsupervised approaches. Our PRLVS ranks first and leads by a large margin, by 0.7%-17.5% on SumMe and 6.6%-19.5% on TVSum, respectively. Compared with the objective of ACGAN [44], our PRLVS is similar in the concept that the key of video summary is capturing global dependencies. ACGAN uses multi-head self-attention to pursue global temporal relationships. In contrast, our PRLVS provides hierarchical reward functions adapting the fine-grained selection from a global perspective.

4.4.2. Comparison in terms of rank correlation coefficients

Table 3 reports the results in terms of τ and ρ . As the results of some competitors are not reported in the original paper, we compare our method with five supervised methods (i.e., dpLSTM, DSNet-AB, DSNet-AF, RSGN, and iPTNet) and two unsupervised methods (i.e., DR-DSN and RSGN_{uns}). Our PRLVS has the closest result to that of human assessment and works considerably better than competitors on TVSum.

4.4.3. Comparison with PRLVS's variants

We also compare PRLVS with its variants in Table 3: 1) **PRLVS-a**, the model trained with database augmented by OVP and YouTube databases; 2) **PRLVS-s**, the model trained with a supervised reward; 3) **PRLVS-f**, the model trained with a flat policy.

Comparison with PRLVS-a. PRLVS-a is an upgraded PRLVS augmenting training database with the OVP and YouTube databases. During each epoch, we augment the training set by randomly select 25% of the size of the original training set from OVP and YouTube. The result shows that the data augmentation contributes to improving the performance on both SumMe and TVSum. This finding validates the influence of the additional data on our PRLVS.

Comparison with PRLVS-s. In PRLVS-s, we replace the unsupervised reward r^h with a supervised reward which calculates the F-score between generated summary and the ground truth. The comparison reports the effect of unsupervised reward r^h . It is interesting to see that the summarization performance of our unsupervised PRLVS is even superior to that of the supervised version, PRLVS-s, on TVSum. This finding validates the effect of our unsupervised reward.

Comparison with PRLVS-f. In PRLVS-f, we flatten the hierarchical policy with an unsupervised reward by simply adding r^h and r^v . The comparison reports the effect of our "T"-type policy. The result shows that PRLVS performs better than PRLVS-f on both SumMe and TVSum. This finding validates the influence of the hierarchical policy on our PRLVS. Although the structure of PRLVS-f is analogous to that of DR-DSN, PRLVS-f adopts our reward measuring the amount of information and achieves better than DR-DSN (See 'US' entry in Table 2). This study further confirms the efficacy of our reward.

4.4.4. Comparison in the augmented and transfer settings

Augmented (A) and Transfer (T) settings for video summarization are scenarios besides the standard setting, with extra data (i.e., OVP and YouTube). In total, we have three settings:

- (1) Canonical (C): The standard setting where the training and testing sets are from the same database.
- (2) Augmented (A): We randomly use 20% of the given database as the testing set, and the rest 80% with the other three databases as the training set.
- (3) Transfer (T): On the given database, we test a model trained on the combination of the other three databases.

The results in Table 5 show that in the Augmented setting, our PRLVS is improved on SumMe while it is impaired on TVSum. Note that the model in the Augmented setting can be regarded as the extension of our PRLVS-a (See Entry 'PRLVS-a' in Table 4).

Table 5

Quantitative comparison with methods in terms of F-score in the canonical (C), Augmented (A) and Transfer (T) settings on SumMe and TVSum databases. The best performance in unsupervised methods is in **bold** and the second-best performance is underlined.

Supervision	databases	SumMe			TVSum		
	Method	C	A	T	C	A	T
FS	dppLSTM	38.6	42.9	41.8	54.7	59.6	58.7
	DSNet-AB	50.2	50.7	46.5	62.1	63.9	59.4
	DSNet-AF	51.2	53.3	47.6	61.9	62.2	58.0
	RSGN	45.0	45.7	44.0	60.1	61.1	60.0
	iPTNet	54.5	56.9	49.2	63.4	64.2	59.8
US	DR-DSN	41.4	42.8	<u>42.4</u>	57.6	58.4	<u>57.8</u>
	RSGN _{uns}	<u>42.3</u>	<u>43.6</u>	41.2	<u>58.0</u>	<u>59.1</u>	59.7
	PRLVS (ours)	46.3	49.7	47.6	63.0	59.2	57.0

Table 6

Dependency on summary size K .

K	SumMe			TVSum		
	F-score	τ	ρ	F-score	τ	ρ
10	46.27	0.07	0.09	62.98	0.08	0.13
15	47.00	0.04	0.06	61.33	0.10	0.11
20	46.26	0.06	0.08	62.14	0.10	0.11

PRLVS-a outperforms PRLVS on TVSum. Compared to PRLVS-a, the main difference is adding SumMe besides OVP and YouTube. Therefore, we further perform the cross-database analysis on SumMe and TVSum in Section 4.6.3. Similarly, in the Transfer setting, our PRLVS is improved on SumMe while it is impaired on TVSum. Note that most of the reported models degraded, inspiring us to explore domain adaptation in video summarization in the future.

4.5. Visualization

In addition to the quantitative comparison, we give a visual comparison result on a test video in TVSum (Fig. 3). The orange bins are shots we converse from key frames, following [8,19]. The original video is about a motorcycle stunt. We compare the summary frames picked by DR-DSN and our PRLVS. As seen, thanks to the proposed hierarchical selection strategy, our result embodies two characteristics: 1) skips camera lens containing half a motorcycle by anchoring representative frames with the horizontal policy, while DR-DSN does not (e.g., the 5th and 6th visual frames picked by DR-DSN); 2) organizes the whole summary combining close-up shots and long shots, and composing dynamic shots and static shots with the vertical policy. In general, our PRLVS selects most of the frames that show the details of the motorcycle stunt. In contrast, DR-DSN also picks some irrelevant frames (e.g., 4th frame in picked frames). A higher F-score and correlation coefficient are therefore achieved by using our PRLVS. We propose qualitative video summarization results of two exemplar videos in Fig. 4. Fig. 4(a) shows an exemplar from SumMe, and Fig. 4(b) shows an exemplar from TVSum.

We also draw correlation curves produced for videos from TVSum database. The red lines illustrate the curve for each human annotator, and the black dashed line is the expectation for random importance scores. The blue curves show the results of our PRLVS. According to the experimental setting in [37]. The light-blue color illustrates the area where the curve lies on the upper bound of the light-blue area if the score rankings are in the same order as the reference and on the lower bound otherwise.

As illustrated in Fig. 5, most human annotators obtain a curve that is above the random baseline. The visualized curve shows that the importance score curve generated by our PRLVS is positively correlated with human annotations.

4.6. Ablation study

4.6.1. Dependency on the summary size K

Table 6 reports the results of our PRLVS on different values of the size of intermediate summary K . K is a hyper-parameter controlling the frame selection in our PRLVS, which is also a significant factor affecting the performance. We set K to 10, 15, 20 and evaluate PRLVS on both TVSum and SumMe. The result shows that the performance of our PRLVS has low dependence on K .

4.6.2. Dependency on the batch size B

Table 7 reports the results of our PRLVS on different values of batch size B . We set B to 1, 8, 16, 32, 64 and evaluate PRLVS on both TVSum and SumMe. The result shows that the performance of our PRLVS has low dependence on B . When in a setting of $B = 1$, our PRLVS needs fewer computation resources than in other settings, making training on limited conditions possible. Thus, we report our main results in the setting of $B = 1$.

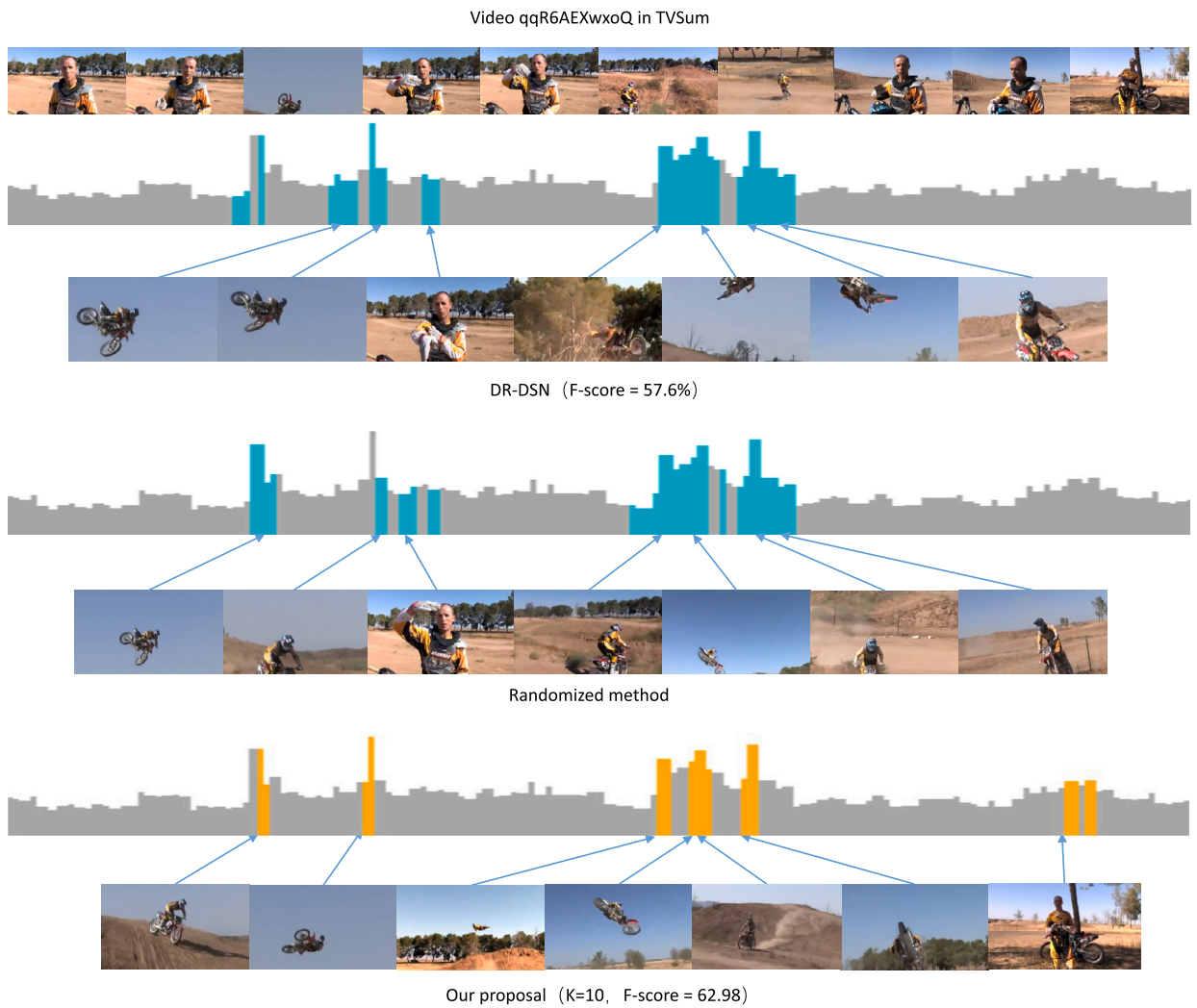


Fig. 3. Exemplar video summary generated by our proposal and DR-DSN, along with the ground-truth importance scores (gray background). Orange bins represent the summaries selected by our method, while Cyan bins stand for the summaries selected by DR-DSN.

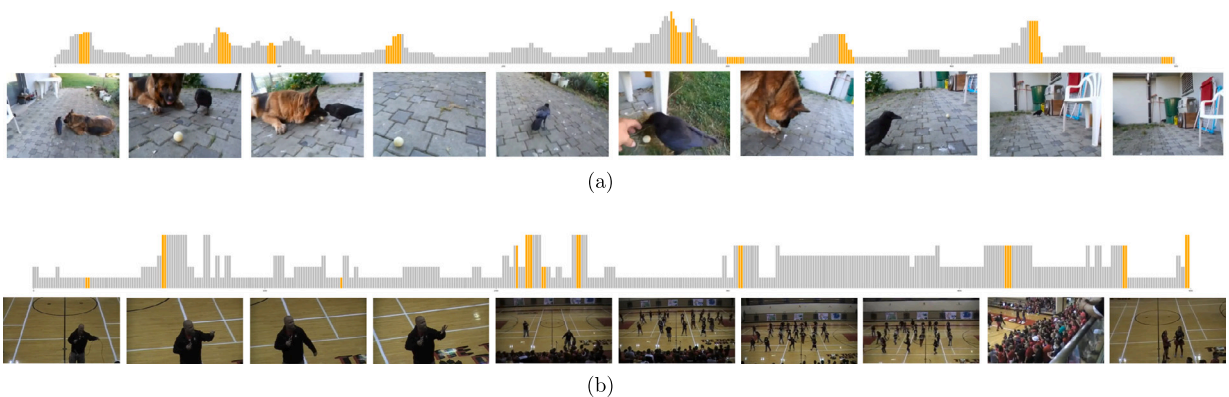


Fig. 4. Two examples of video summary generated by our PRLVS from SumMe (a) and TVSum (b). Orange bins represent the summaries selected by our method, while gray bins stand for the ground-truth importance scores.

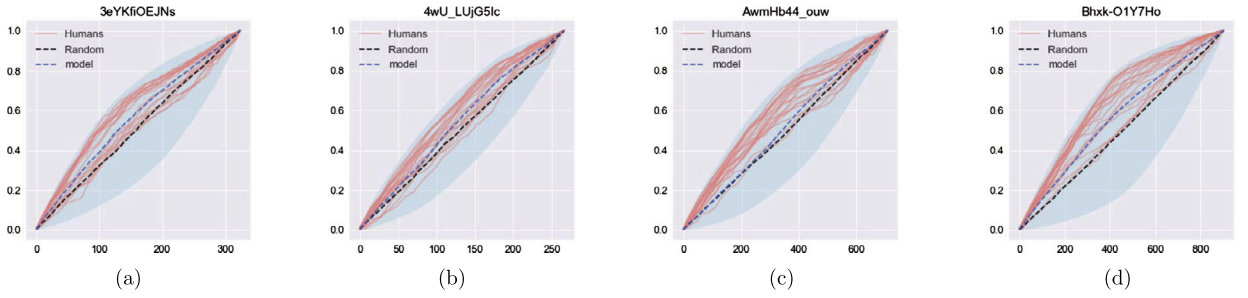


Fig. 5. Example correlation curves produced for four videos from TVSum database (3eYKfiOEJNs, 4wU_LUjG5lc, AwmHb44_ouw, and Bhxk-O1Y7Ho are video ids). The red lines represent correlation curves for each human annotator, and the black dashed line is the expectation for random importance scores. The blue and green curves show the corresponding results to our PRLVS.

Table 7
Dependency on batch size B .

batch size	SumMe			TVSum		
	F-score	τ	ρ	F-score	τ	ρ
1	46.27	0.07	0.09	62.98	0.08	0.13
8	46.05	0.08	0.09	63.18	0.08	0.13
16	46.13	0.08	0.09	63.11	0.09	0.14
32	46.09	0.08	0.09	63.07	0.09	0.13
64	46.15	0.08	0.09	63.03	0.09	0.14

Table 8
Dependency on the training database.

$\text{tr}^a \backslash \text{t}^b$	SumMe			TVSum		
	F-score	τ	ρ	F-score	τ	ρ
SumMe	46.3	0.07	0.09	62.4	0.14	0.18
TVSum	45.7	0.03	0.07	63.0	0.13	0.08

^a “tr” is short for training.

^b “t” is short for testing.

4.6.3. Dependency on the training database

We propose a cross-database validation on SumMe and TVSum, to perform a robust evaluation of our PRLVS. The key idea is to train our model on one database and test it on another one. The construction of SumMe and TVSum utilize different sources, resulting in different distributions. Thus, the cross-database validation can verify whether our method generalizes well to new distributions. Table 8 reports the results of a cross-database experiment. The F-score of our PRLVS only drops 1.30% and 0.95% in cross-database validation on SumMe and TVSum, respectively. We can find that the difference among training databases has a limited influence on performance. This finding validates the generalization ability of our PRLVS.

Interestingly, we find that our PRLVS achieves higher rank correlation coefficients in cross-database validation on TVSum. A possible explanation is that SumMe is more sparse than TVSum in frame-level annotations (Non-zero important score proportion: 12.96% to 100.00%), the model trained on SumMe prefers learning ordinal information to learning frame selection binary value [45]. F-score concentrates on frame-level selection binary value by counting whether some key frames are selected. At the same time, correlation coefficients attend to the overall ordinal trend by calculating the similarity of important score curves. Thus, our PRLVS achieves higher rank correlation coefficients while lower F-score in cross-database validation on TVSum. In sum, the model trained on SumMe has better generalization than the model trained on TVSum. This finding is similar to the principle of Dropout [46], which Occam’s Razor can also explain.

5. Conclusion

In this paper, we propose a progressive reinforcement learning (PRL) strategy to address video summarization. To implement the strategy by simulating the “T”-type human thinking paradigm, we design a horizontal policy and a vertical policy to alternatively reason a picking action to select key frames. The two policies are constructed hierarchically to provide more information than flat strategies with sparse rewards, ensuring the integrity of generated summaries. Our PRLVS optimizes this hierarchical strategy in an unsupervised manner, exploiting global long-term temporal information and local short-term dynamics.

Extensive experiments show that our approach achieves competitive performance over comparisons known to us on benchmarks TVSum and SumMe. In particular, our PRLVS achieves comparable performance close to state-of-the-art supervised methods.

CRediT authorship contribution statement

Guolong Wang: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Xun Wu:** Conceptualization, Formal analysis, Resources, Software, Writing – original draft, Writing – review & editing. **Junchi Yan:** Conceptualization, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This work was partially supported by the Fundamental Research Funds for the Central Universities in UIBE (22QN01) and Shanghai Municipal Science and Technology Major Project (2021SHZDZX0102).

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